

Practice Quiz: Reasoning About Results

Part A: Multiple Choice

1. Bayesian updating in prototype testing means: A) Rejecting the prototype after any negative test result B) Combining prior beliefs with test evidence proportionally to form updated beliefs C) Ignoring prior beliefs and relying solely on test data D) Using only statistical software to analyze results
2. An inventor who treats random test variation as a meaningful signal is likely to: A) Build a better prototype B) Over-correct the design, increasing variability rather than reducing it C) Reduce manufacturing costs D) Discover new failure modes
3. When test results contradict the generative model, the three possible responses (from least to most disruptive) are: A) Ignore, deny, abandon B) Parameter updating, structure revision, model replacement C) Test again, hire an expert, start over D) Celebrate, investigate, panic
4. Confirmation bias in prototype testing occurs when: A) Multiple testers confirm the same result B) The inventor interprets ambiguous evidence as supporting existing beliefs due to asymmetric likelihood weighting C) The test protocol confirms correct procedures were followed D) Previous results are confirmed by new data
5. The Analysis of Competing Hypotheses (ACH) method improves reasoning by: A) Eliminating the need for testing B) Forcing the inventor to evaluate multiple explanations against the evidence, not just the preferred one C) Using only quantitative data D) Requiring consensus among all team members
6. Pre-registration of test hypotheses and criteria helps prevent: A) Test equipment failure B) Post-hoc rationalization and cherry-picking of supportive results C) Budget overruns D) Prototype damage during testing
7. The Challenger disaster is an example of: A) A mechanical failure that could not have been predicted B) Reasoning failures including base rate neglect, burden of proof

inversion, and motivated reasoning under schedule pressure C) Successful risk management that was overruled by politicians D) A failure of manufacturing quality control

Part B: Short Answer

1. You test a new adhesive formulation. Three tests show bond strength of 150 psi, 145 psi, and 148 psi. A fourth test shows 95 psi. Should you treat the fourth result as signal or noise? Explain your reasoning using the concepts of repeatability, precision, and common vs. special cause variation.
2. An inventor receives two pieces of evidence about their prototype: (a) quantitative testing shows it meets performance specifications, and (b) qualitative user feedback says "I wouldn't buy this." These results appear contradictory. Using the concept of model structure vs. model parameters, explain how both can be correct and what the inventor should do.
3. You have been working on an invention for 18 months. The latest round of testing produces results that suggest your fundamental approach may be wrong. Describe the sunk cost bias that might affect your reasoning, explain why it is irrational from a Bayesian perspective, and propose a structured protocol for making the continue/pivot/abandon decision.